FIITJEE Students From Classroom / Integrated School Programs have secured to 2 Zonal, 6 State & 18 City Topper Ranks. 33 in Top 100, 78 in Top 200 and 205 in Top 500 All India Ranks bagged by FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

ALL INDIA TEST SERIES

FIITJEE

JEE (Advanced)-2016

FULL TEST – I

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 246**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-B & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section – A (01 – 07)** contains 7 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

   Section – A (08 – 11) contains 4 multiple choice questions which have one or more than one correct answer. Each question carries **+3 marks** for correct answer. There is no negative marking.

   Section-A (12 – 16) contains 2 paragraphs. Based upon paragraph, 2 and 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

2. **Section – B (01 – 02)** contains 2 Matrix Match Type questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+8 marks** for all correct answer. For each correct row **+2 marks** will be awarded. There may be one or more than one correct matching. No marks will be given for any wrong matching in any question. There is no negative marking.

3. **Section-C (01 – 06)** contains 6 Numerical based questions with answers as numerical value from **0 to 9** and each question carries **+3 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

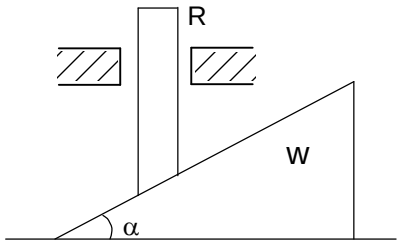
(Straight Objective Type)

This section contains **7 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** option **is correct**.

1. A flat coil of area A and n turns is placed at the centre of a ring of radius r (r^2 >> A) and resistance R. The two are co-planar when current in the coil increases from zero to i, the total charge circulating in the ring is

(A) $\frac{\mu_0 nAi}{2rR}$ (B) $\frac{\mu_0 nAi}{r^2R}$ (C) $\frac{\mu_0 nAi}{2\pi R}$ (D) $\frac{\mu_0 n^2 i}{4\pi rR}$

2. In the arrangement shown, all surfaces are frictionless. The rod R is constrained to move vertically. The vertical acceleration of R is a_1 and the horizontal acceleration of the wedge w is a_2. The ratio $\frac{a_1}{a_2}$ is equal to



(A) $\tan\alpha$ (B) $\cot\alpha$
(C) $\sin\alpha$ (D) $\cos\alpha$

3. An engine whistling at a constant frequency n_0 and moving with a constant velocity goes past a stationary observer. As the engine crosses him, the frequency of the sound heard by him changes by a factor f. The actual difference in the frequency of the sound heard by him before and after the engine across him is

(A) $\frac{1}{2}n_0\left(1-f^2\right)$ (B) $\frac{1}{2}n_0\left(\frac{1-f^2}{f}\right)$ (C) $n_0\left(\frac{1-f}{1+f}\right)$ (D) $\frac{1}{2}n_0\left(\frac{1-f}{1+f}\right)$

4. A ball of density ρ_0 falls from rest from point P onto the surface of a liquid of density ρ in time T. It enters the liquid stops moves up and returns to P in a total time 3T. Neglect viscosity, surface tension and splashing. The ratio $\frac{\rho}{\rho_0}$ is equal to

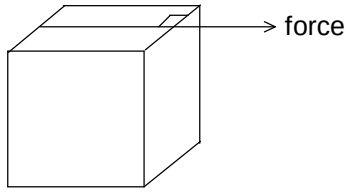
(A) 1.5 (B) 2 (C) 3 (D) 4

Space for Rough work

5. When an object is placed in front of a concave mirror of focal length f, a virtual image is produced with a magnification of 2. To obtain a real image with a magnification of 2. The object has to be moved by a distance equal to

(A) $\frac{f}{2}$ (B) $\frac{2}{5f}$ (C) f (D) $\frac{3}{2}f$

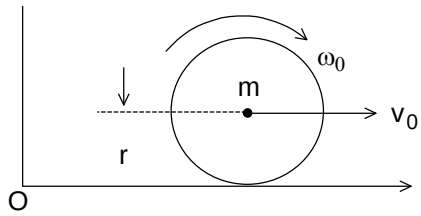
6. A solid cube is placed on a horizontal surface. The coefficient of friction between them is μ, where $\mu < \frac{1}{2}$. A variable horizontal force is applied on the cube's upper face, perpendicular to one edge and passing through the mid-point of that edge. The maximum acceleration with which it can move without toppling is



(A) μg (B) $2\mu g$

(C) $\frac{g}{2}(1-2\mu)$ (D) $\left(\mu+\frac{1}{2}\right)$

7. A spherical body of mass m, radius r and moment of inertia I about its centre moves along the x-axis. Its centre of mass moves with velocity = v_0 and it rotates about its centre of mass with angular velocity ω_0. Let $L_0 = I\omega_0 + mv_0 r$. The angular momentum of the body about the origin O is



(A) L_0, only if $v_0 = r\omega_0$ (B) greater than L_0 if $v_0 > \omega_0 r$

(C) less than L_0, if $v_0 > \omega_0 r$ (D) L_0 for all values of ω_0 and v_0

Multiple Correct Choice Type

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

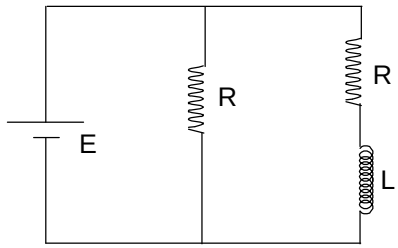
8. The temperature of a solid object is observed to be constant during a period. In this period
(A) heat may have been supplied to the body
(B) heat may have been extracted from the body
(C) no heat is supplied to the body
(D) no heat is exterted from the body

Space for Rough work

9. The electric potential decreases uniformly from 120 V to 80 V as one moves on the axis from x = –1 cm to x = 1 cm. The electric field at the origin
(A) must be equal to 20 V cm^{-1} (B) may be equal to 20 V cm^{-1}
(C) may be greater than 20 V cm^{-1} (D) may be less than 20 V cm^{-1}

10. A bag of mass M hangs by a long thread and a bullet (mass m) comes horizontally with velocity v and gets caught in the bag. Then for the combined system (bag + bullet)
(A) momentum is $\frac{mMv}{(M+m)}$ (B) KE = $\frac{1}{2}Mv^2$
(C) momentum is mv (D) KE is $\frac{m^2v^2}{2(M+m)}$

11. In the circuit diagram shown
(A) time constant is $\frac{L}{R}$
(B) time constant is $\frac{2L}{R}$
(C) steady state current in inductor is $\frac{2E}{R}$
(D) steady state current in inductor is $\frac{E}{R}$

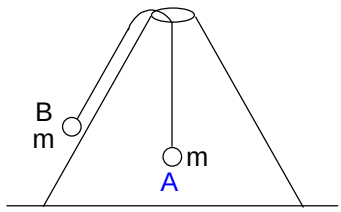


Comprehension Type

This section contains **2 paragraphs.** Based upon the first paragraph 2 multiple choice questions and based upon the second paragraph **3 multiple choice questions** have to be answered. Each of these questions has four choices A), B), C) and D) out of **WHICH ONLY ONE IS CORRECT.**

Paragraph For Question No.12 and 13

The base of a hollow right cone of semi vertical angle 30° is fixed to a horizontal plane. Two particle each of mass m are connected by a light inextensible string which passes through a small hole in the vertex of the cone. One particle A hangs at rest inside the cone. The other particle B moves on the outer smooth surface of the cone at a distance ℓ from vertex in a horizontal circle with centre at A. Neglecting friction, now answer the following.



12. The tension in the string is
(A) $\frac{\sqrt{3}}{2}mg$ (B) $\frac{1}{2}mg$ (C) $\frac{2}{\sqrt{3}}mg$ (D) none of these

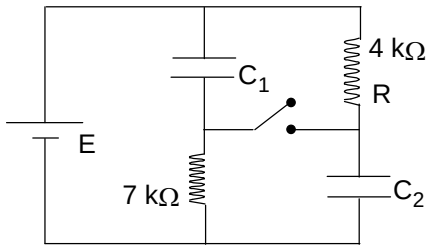
Space for Rough work

13. The angular velocity of B is

(A) $\sqrt{\frac{2g}{\ell}\left(2\sqrt{3}-1\right)}$ (B) $\sqrt{\frac{2g}{\ell}\left(\sqrt{3}-1\right)}$ (C) $\sqrt{\frac{2g}{\ell}\left(2-\sqrt{3}\right)}$ (D) $\sqrt{\frac{g}{\ell}\left(2-\sqrt{3}\right)}$

Paragraph For Question No.14 and 16

The switch S has been closed for long time and the electric circuit shown carries a steady current. Let C_1 = 3μF, C_2 = 6μF, R_1 = 4 kΩ and R_2 = 7.0 kΩ. The power dissipated in R_2 is 2.8 W.



14. The power dissipated to the resistor R_1 is

(A) 2.8 W (B) 1.6 W (C) 4.9 W (D) 0

15. The charge on capacitance C_1 and C_2 are respectively

(A) 940 μC, 940 μC (B) 440 μC, 440 μC (C) 240 μC, 840 μC (D) 840 μC, 240 μC

16. Long time after switch is opened the charge on C_1 is

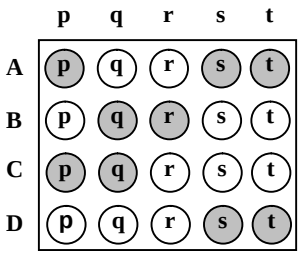
(A) zero (B) 420 μC (C) 240 μC (D) 660 μC

Space for Rough work

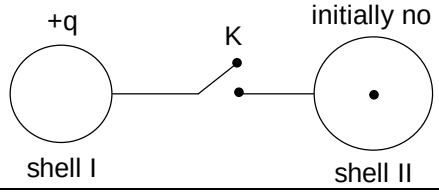
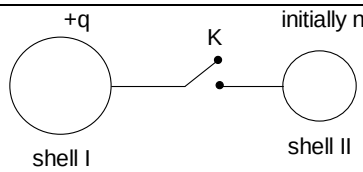
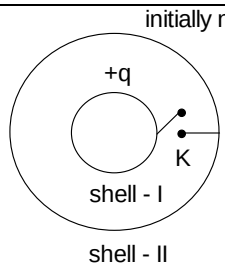
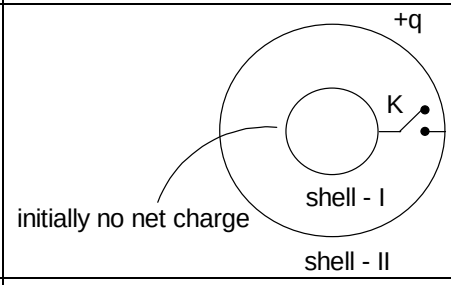
SECTION-B
(Matrix-Match Type)

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



1. Column – I gives certain situation involving two thin conducting shells connected by a conducting wire via a key K. In all situation one sphere has net charge +q and other sphere has no net charge. After the key K is pressed column – II gives some resulting effect. Match the figures in column with the statements in column – II.

Column – I		Column – II	
(A)		(p)	Charge flow through the connecting wire
(B)		(q)	Potential energy of system of spheres decreases
(C)		(r)	No heat is produced
(D)		(s)	The sphere I has no charge after equilibrium is reached
		(t)	Potential energy of system of sphere increase

Space for Rough work

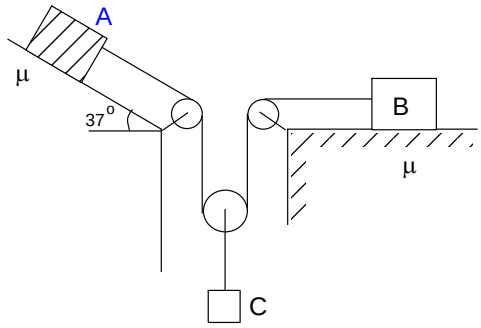
2. A particle of mass 2 kg is moving on straight line under the action of force F = (8 – 2x)N. The particle is released at rest from x = 6m. For the sub-sequent motion match the following (All the values in the right column are in there SI units).

Column – I		Column – II	
(A)	Equilibrium position is at x	(p)	$\frac{\pi}{4}$
(B)	Amplitude of SHM is	(q)	$\frac{\pi}{2}$
(C)	Time taken to go directly from x = 2 to x = 4	(r)	4
(D)	Energy of SHM is	(s)	6
		(t)	2

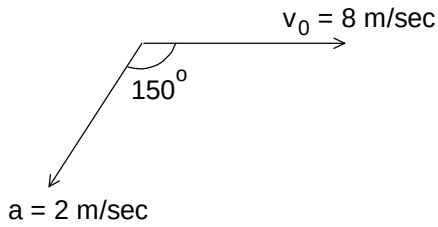
SECTION –C
Integer Answer Type

This section contains **6 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. In the arrangement shown in figure, pulley are light and frictionless, threads are inextensible and mass of blocks A, B and C are m_1 = 5 kg, m_2 = 4 kg and m_3 = 2.5 kg respectively and co-efficient of friction for both the planes is μ = 0.50. Calculate the acceleration of block A (in m/sec), when the system is released from rest. (g = 10 m/sec)

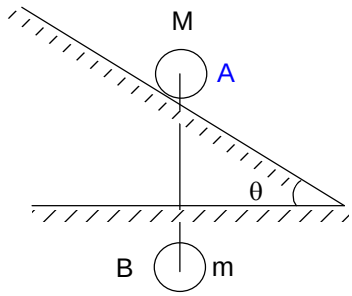


2. The figure shows the velocity and acceleration of a point like body at the initial moment of its motion. The acceleration vector of the body remains constant. The minimum radius of curvature of trajectory of the body is

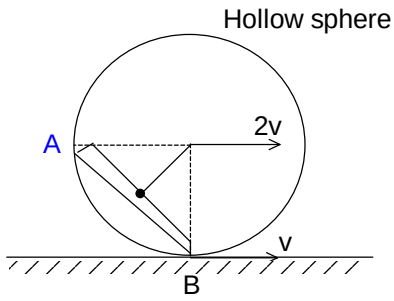


Space for Rough work

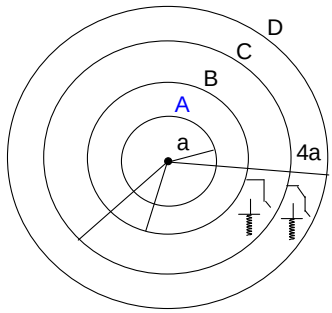
3. A disc 'A' of mass M is placed at rest on the smooth inclined surface of inclination θ. A ball B of mass m is suspended vertically from the centre of the disc A by a light inextensible string of light ℓ as shown in the figure. If the acceleration of the disc B immediately after the system is released from rest is $\frac{(M+km)g\sin^2\theta}{M+m\sin^2\theta}$. Find k



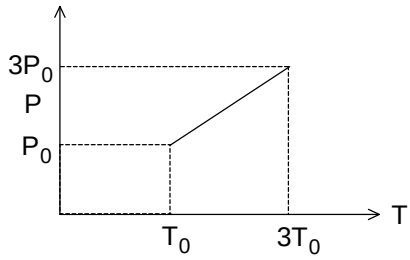
4. A uniform rod AB of mass M and length $R\sqrt{2}$ is moving in a vertical plane inside a hollow sphere of radius R. The sphere is rolling on a fixed horizontal surface without slipping with velocity of its centre of mass 2v, when the end B is at the lowest position, its speed is found to be v as shown in the figure. If the kinetic energy of rod at this instant is $\frac{4}{k}Mv^2$. Find k.



5. In the figure A, B, C and D are four concentric spheres of radius a, 2a, 3a and 4a and contain, charges +3q, +6q, -6q and +12q respectively. In the beginning switches S_1 and S_2 are open. After earthing the sphere C Δq_1 amount of charge flows from the sphere C to the earth. If the sphere B is also earth, then Δq_2 amount of charge flows from the sphere B to the earth, find the value of $\frac{|\Delta q_1|}{|\Delta q_2|}$.



6. Two moles of a monoatomic ideal gas undergoes a process AB as shown in the figure. Then find the work done during the process in Joule.



Space for Rough work

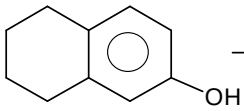
Chemistry

PART – II

SECTION – A

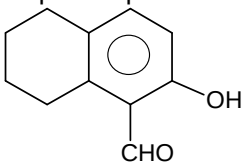
Straight Objective Type

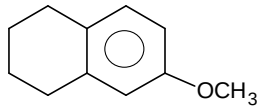
This section contains **7 multiple choice** questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONLY ONE** option is correct.

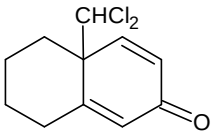
1. 

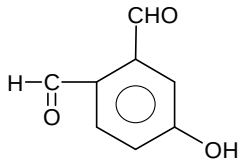
A (Expected) + B (Unexpected)

The unexpected product B is

(A) 

(B) 

(C) 

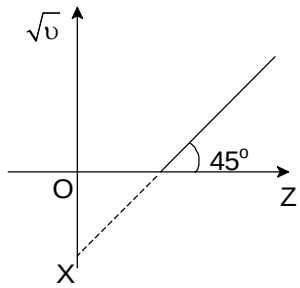
(D) 

2. $I_2(s) \mid I^-$ (0.1 M) half cell is connected to a H^+ (aq.) | H_2(1 bar) | Pt half cell and emf is found to be 0.7714 V. If $E^o_{I_2|I^-} = 0.535$ V, find the pH of $H^+ \mid H_2$ half cell.

(A) 1 (B) 3 (C) 5 (D) 7

3. In the graph between $\sqrt{\upsilon}$ and Z for the Mosley's equation, the intercept OX is -1 on $\sqrt{\upsilon}$ axis. What is the frequency when atomic number (Z) is 51?

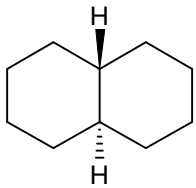
(A) 50 Hz (B) 100 Hz
(C) 2500 Hz (D) None

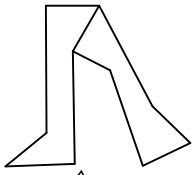


Space for Rough work

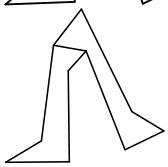
4. Sulphur reacts with chlorine in 1 : 2 ratio and forms X. Hydrolysis of X gives a sulphur compound Y. What is the hybridization state of central atom in the compound Y?
(A) sp^3 (B) sp^2 (C) sp (D) dsp^2

5. Find out correct representation of trans - decaline



(A) 

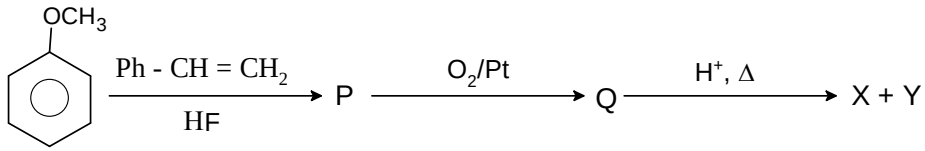
(B) 

(C) 

(D) None of these

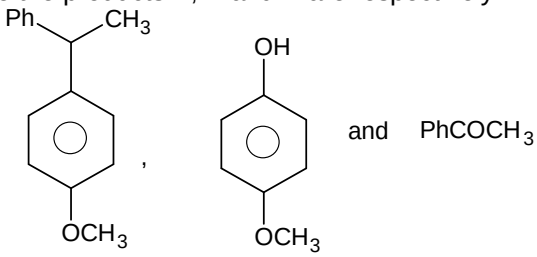
6. 50 ml of a solution containing 10^{-3} mol of Ag^+ is mixed with 50 ml of a 0.1 M HCl solution. How much $[Ag^+]$ remains in solution? Given: K_{sp} of AgCl = 10^{-10}
(A) 2.5×10^{-9} M (B) 2.5×10^{-7}M (C) 2.5×10^{-8}M (D) 2.5×10^{-10}M

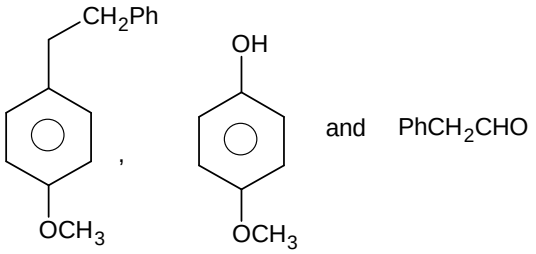
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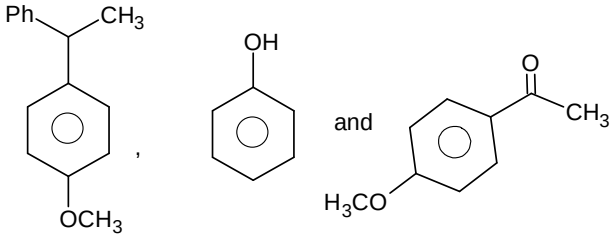
7. 

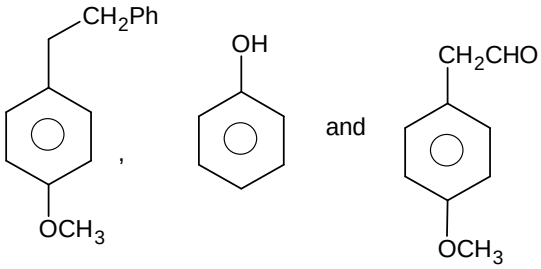
Where X is a compound that forms azodye with benzene diazonium chloride in faintly basic medium.

Hence the products P, X and Y are respectively.

(A) 

(B) 

(C) 

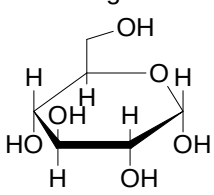
(D) 

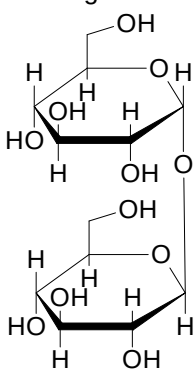
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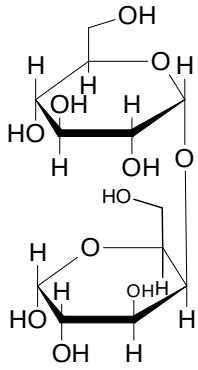
Multiple Correct Choice Type

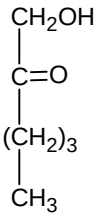
This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

8. Some organic compounds are given below:


(I)


(II)


(III)


(IV)

Select the correct statement(s) about the above compounds from the following:
(A) Compounds (I) and (III) will reduce Fehling's solution and will also mutarotate
(B) Compound (IV) will reduce Tollen's reagent but not mutarotate
(C) Compound (II) will not be reducing but mutarotating
(D) The product(s) of the acid catalysed hydrolysis of both (II) and (III) will be reducing but not mutarotating

9. Which of the following solutions show lowering of vopour pressure on mixing?
(A) Aniline and cyclohexane
(B) Chloroform and acetone
(C) Ethyl alcohol and pentane
(D) Dimethyl ether and chloroform

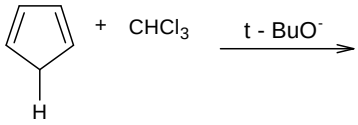
10. HI cannot be prepared by the action of conc. H_2SO_4 on KI because
(A) HI is stronger acid than H_2SO_4
(B) H_2SO_4 is an oxidizing agent
(C) HI is a strong reducing agent
(D) H_2SO_4 is stronger acid than HI

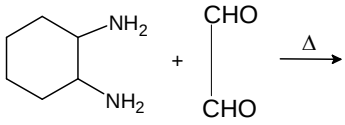
Space for Rough work

11. Which of the following will produce aromatic compound as a product?

(A) $CH_3 - C \equiv CH \xrightarrow[\Delta]{\text{Redhot Fe tube}}$

(B) $n - C_6H_{14} \xrightarrow[Cr_2O_3/600^oC]{Al_2O_3}$

(C) 

(D) 

Comprehension Type

This section contains **2 paragraphs.** Based upon the first paragraph 2 multiple choice questions and based upon the second paragraph **3 multiple choice questions** have to be answered. Each of these questions has four choices A), B), C) and D) out of **WHICH ONLY ONE IS CORRECT.**

Paragraph For Question No.12 and 13

A white compound (A) on strong heating decomposes to produce two products (B) and (C). (B) on reaction with white phosphorus produces (D), which is a strong dehydrating agent. (D) on reaction with perchloric acid converts it to its anhydride.

12. The compound (A) is

(A) NH_4NO_2 (B) NH_4ClO_4 (C) $(NH_4)_2Cr_2O_7$ (D) NH_4NO_3

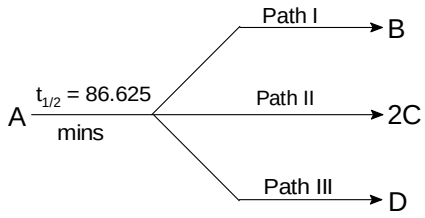
13. The product/s obtained on hydrolysis of (D) is

(A) Mixture of H_3PO_3 and H_3PO_4
(B) Only PH_3
(C) Only H_3PO_4
(D) Mixture of H_3PO_3 and H_3PO_2

Space for Rough work

Paragraph For Question No.14 and 16

'A' is a substance that converts into B, C and D by three first order parallel paths simultaneously according to the following stoichiometry



The partial $t_{1/2}$ of A along path I and path II are 173.25 min and 346.5 min respectively. The energies of activation of the reaction along path I, path II and path III are 40, 60 and 80 kJ/mol respectively.

14. The percent distribution of C in the product mixture B, C and D at any time is equal to
(A) 20 (B) 60 (C) 80 (D) 40

15. The initial rate of consumption of A and the sum of the initial rate of formation of B, C and D are respectively, taking [A] = 0.25 M, equal to
(A) $2\times10^{-3}\,mol L^{-1}min^{-1}$ and $2.5\times10^{-3}\,mol\,L^{-1}min^{-1}$
(B) $2\times10^{-3}\,mol\,L^{-1}min^{-1}$ and $2\times10^{-3}\,mol\,L^{-1}min^{-1}$
(C) $2.5\times10^{-3}\,mol L^{-1}min^{-1}$ and $3\times10^{-3}\,mol\,L^{-1}min^{-1}$
(D) $4\times10^{-3}\,mol L^{-1}min^{-1}$ and $2\times10^{-3}\,mol\,L^{-1}min^{-1}$

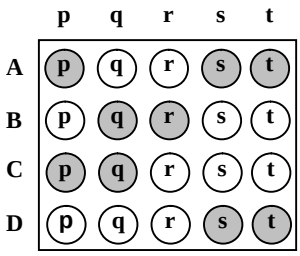
16. The overall energy of activation of A along all the three parallel path is equal to
(A) 52 kJ/mol (B) 60 kJ/mol (C) 55 kJ/mol (D) 80 kJ/mol

Space for Rough work

SECTION-B
(Matrix-Match Type)

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

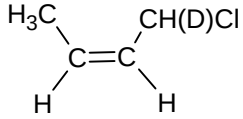
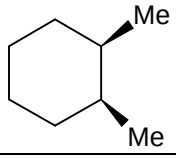
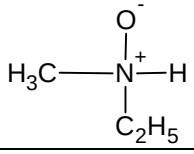
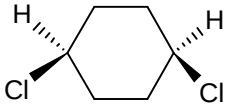


1. Match the following:

Column – I		Column - II	
(A)	$XeF_4 + H_2O(\ell) \longrightarrow$	(p)	Shows disproportionation reaction
(B)	$P_4(s) + OH^-(aq) + H_2O(\ell) \longrightarrow$	(q)	One of the product is a monobasic Lewis acid
(C)	$B_2H_6 + H_2O(\ell) \longrightarrow$	(r)	One/two of the products is/are gas(es)
(D)	$Br_2(g) + k_2CO_3(aq) \longrightarrow$	(s)	One of the product has trigonal pyramidal structure
		(t)	One of the gaseous product is used as Holme's signal in sea

Space for Rough work

2. Match the following:

Column – I		Column - II	
(A)		(p	Optically active
(B)		(q	Optically inactive
(C)		(r)	Plane of symmetry
(D)		(s)	Number of chiral centre = 1
		(t)	Number of stereocentres = 2

SECTION –C
Integer Answer Type

This section contains **6 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. Number of nuclear particles (Projectiles + ejectiles) involved in the conversion of the nuclide $_{92}U^{238}$ into $_{94}Pu^{239}$ by neutron – capture method is .....

2. A crystal is made up of atoms X, Y and Z. Atoms X are in FCC packing. Y occupies all octahedral voids and Z occupies all tetrahedral voids. If all the atoms along two body diagonals are removed, the ratio of sum of effective number of atoms of Y and Z to the effective number of atoms of X is a : 1, what is the value of a?

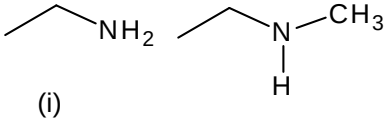
Space for Rough work

3. How many aldol products may be formed by the reaction of CH_3CH_2CHO with $CH_3COCH_2CH_3$?

4. 5.6 L an unknown gas at NTP requires 12.5 calories to rise its temperature by 10^oC at constant volume. What is the atomicity of the gas?

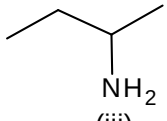
5. After electrolysis of a NaCl solution with inert electrodes for a certain period of time, 60 ml of the solution was left which was found to be 1 N in NaOH. During the same time 31.75 g of Cu was deposited in a Cu voltameter in series with the electrolytic cell. Calculate the percentage of the theoretical yield of the sodium hydroxide obtained.

6. Of the following amines how many can be separated by Hoffmann's mustard oil reaction?

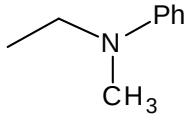


(i)

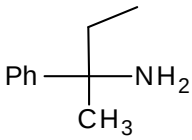
(ii)



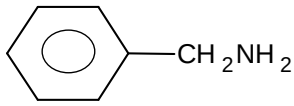
(iii)



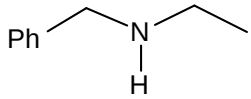
(iv)



(v)



(vi)



(vii)

Space for Rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains 7 multiple choice questions numbered 1 to 7. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. If $f(x)=\frac{\int_0^n [x]dx}{\int_0^n \{x\}dx}$, then $\sum_{n=1}^{n} f(x)$ is equal to (where [•] and {•} denotes GIF and fraction parts of x), $n \in N$

(A) $n^2 - 1$ (B) $\frac{n^2-n}{2}$ (C) $\frac{n(n+1)}{2}$ (D) $n^2 + 1$

2. The solution of the differential equation $y'y''' = 3(y'')^2$ is

(A) $x = A_1y^2 + A_2y + A_3$ (B) $x = A_1y + A_2$

(C) $x = A_1y^2 + A_2y$ (D) none of these

3. A point on the hypotenuse of a triangle is at distance a and b from the side of the triangle then minimum length of the hypotenuse is

(A) $\left(a^2+b^2\right)^{\frac{1}{2}}$ (B) $\left(a^2+b^2\right)^{\frac{2}{3}}$ (C) $\left(a^{\frac{2}{3}}+b^{\frac{2}{3}}\right)^{\frac{3}{2}}$ (D) $\left(a^{\frac{1}{3}}+b^{\frac{1}{3}}\right)^{\frac{2}{3}}$

4. If the equation $x^4 - 4x^3 + ax^2 + bx + 1 = 0$ has four positive roots then

(A) a = 6, b = –4 (B) a = 5, b = –4 (C) a = 6, b = –3 (D) a = 6, b $\in$ R

5. Find the point P on the line 2x + 3y + 1 = 0 such that |PA – PB| is maximum where A is (2, 0) and B is (0, 2)

(A) (1, –1) (B) (7, –5) (C) (–2, 1) (D) (4, –3)

Space for Rough work

6. Find the coefficient of x^5 in the expansion of $(2 - x + 3x^2)^6$
(A) –5049 (B) 5051 (C) –5052 (D) 5050

7. If $\tan 3\theta + \tan\theta = 2\tan 2\theta$ then θ is equal to ($n \in z$)
(A) $n\pi$ (B) $\frac{n\pi}{4}$ (C) $2n\pi$ (D) none of these

Multiple Correct Choice Type

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct

8. If x, y, z are respectively perpendiculars from the circumcenter to the sides of ΔABC, (a, b, c are usual meanings) then
(A) $\tan A = \frac{a}{2x}$ (B) $\frac{a}{x}+\frac{b}{y}+\frac{c}{z}=\frac{abc}{4xyz}$ (C) $\frac{a}{x}=\tan A$ (D) $\frac{a}{x}+\frac{b}{y}+\frac{c}{z}=\frac{abc}{2xyz}$

9. Which of the following is/are correct
(A) $(\tan x)^{\ln(\sin x)} > (\cot x)^{\ln(\sin x)} \ \forall\ x \in \left(0,\frac{\pi}{4}\right)$ (B) $4^{\ln(\operatorname{cosec} x)} > 5^{\ln(\operatorname{cosec} x)} \ \forall\ x \in \left(0,\frac{\pi}{2}\right)$
(C) $\left(\frac{1}{2}\right)^{\ln(\cos x)} < \left(\frac{1}{3}\right)^{\ln(\cos x)} \ \forall\ x \in \left(0,\frac{\pi}{2}\right)$ (D) none of these

10. The angle between the lines whose direction cosine are connected by the relations ℓ –5m+3n=0 and $7\ell^2 + 5m^2 - 3n^2 = 0$
(A) $\tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (B) $\cos^{-1}\sqrt{\frac{7}{12}}$ (C) $\cos^{-1}\sqrt{\frac{3}{7}}$ (D) $\tan^{-1}\sqrt{\frac{5}{7}}$

11. If $\vec{a},\vec{b}$ and $\vec{c}$ are three mutually perpendicular unit vectors and $\vec{d}$ is a unit vector which makes equal angles with $\vec{a},\vec{b}$ and $\vec{c}$ then the value of $\left|\vec{a}+\vec{b}+\vec{c}+\vec{d}\right|^2$
(A) $4+2\sqrt{3}$ (B) $4-2\sqrt{3}$ (C) $2+\sqrt{5}$ (D) $2-\sqrt{5}$

Space for Rough work

Comprehension Type

This section contains 2 paragraphs. Based upon one of paragraphs 2 multiple choice questions and based on the other paragraph 3 multiple choice questions have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

Paragraph for Question Nos. 12 to 13

PQ is the double ordinate of the parabola $y^2 = 4x$ which passes through the focus S. ΔPQA is an isosceles right angle triangle, where A is on the axis of the parabola. Line PA meets the parabola at C and QA meets the parabola at B.

12. The area of trapezium PBCQ is
(A) 96 sq units (B) 64 sq units (C) 72 sq units (D) 48 sq units

13. The circumradius of trapezium PBCQ is
(A) $6\sqrt{5}$ (B) $3\sqrt{6}$ (C) $2\sqrt{10}$ (D) $5\sqrt{3}$

Paragraph for Question Nos. 14 to 16

Let z be a complex number satisfying $z^2 + 2\alpha z + 1 = 0$ where α is a parameter which can take any real value.

14. The roots of this equation lie on a certain circle if
(A) $-1 < \alpha < 1$ (B) $\alpha > 1$ (C) $\alpha < 1$ (D) none of these

15. One root lies inside the unit circle and one outside if
(A) $-1 < \alpha < 1$ (B) $\alpha > 1$ (C) $\alpha < 1$ (D) none of these

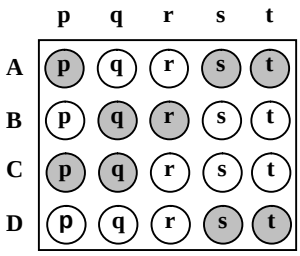
16. For every large value of α the roots are approximately
(A) $-2\alpha, \frac{1}{\alpha}$ (B) $-\alpha, -\frac{1}{\alpha}$ (C) $-2\alpha, -\frac{1}{2\alpha}$ (D) none of these

Space for Rough work

SECTION-B
(Matrix-Match Type)

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



1. A function is defined as $f : \{a_1, a_2, a_3, a_4, a_5, a_6\} \to \{b_1, b_2, b_3\}$

Column – I		Column – II	
(A)	The number of onto functions	(p)	Is divisible by 9
(B)	Number of function in which $f(a_i) \neq b_i$	(q)	Is divisible by 5
(C)	Number of invertible functions	(r)	Is divisible by 4
(D)	Number of many one functions	(s)	Is divisible by 3
		(t)	Not possible

2. Match the following:

Column – I		Column – II	
(A)	$\sin^{-1}(\sin 10)$	(p)	$4\pi - 10$
(B)	$\cos^{-1}(\cos 10)$	(q)	$3\pi - 10$
(C)	$\tan^{-1} x + \tan^{-1}\frac{1}{x}$	(r)	$\frac{\pi}{2}$
(D)	$\sin^{-1}\frac{1}{\sqrt{2}} + \sin^{-1}\left(\frac{\sqrt{2}-1}{\sqrt{6}}\right) + \ldots + \sin^{-1}\left(\frac{\sqrt{n}-\sqrt{n-1}}{\sqrt{n(n+1)}}\right) +$ to ∞	(s)	$-\frac{\pi}{2}$
		(t)	10

Space for Rough work

SECTION – C

(Integer Answer Type)

This section contains **6 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. Two friends have equal number of sons. There are 3 tickets for a cricket match which are to be distributed among the sons. The probability that two tickets go to the sons of one and one tickets goes to the sons of other is $\frac{6}{7}$. Then total number of boys is equal to (sum of son of each friend).

2. A line through the origin meets the circle $x^2 + y^2 = a^2$ at P and the hyperbola $x^2 - y^2 = a^2$ at Q. Then locus of the point of intersections of tangent to the circle at P with the tangent at Q to the hyperbola is the curve $(a^4 + 4y^4)x^2 = a^K$ then K is equal to ..

3. In a ΔABC, $\Delta = \begin{vmatrix} e^{i2A} & e^{-iC} & e^{-iB} \\ e^{-iC} & e^{i2B} & e^{-iA} \\ e^{-iB} & e^{-iA} & e^{i2C} \end{vmatrix}$. Then the value of $|\Delta|$, where $|\bullet|$ denote absolute value.

4. If $\int \frac{1}{\left((x-1)^3(x+2)^5\right)^{\frac{1}{4}}} dx = \frac{P}{q}\left(\frac{x-1}{x+2}\right)^{\frac{1}{4}} + C$. Then P + q is

5. The area of region in first quadrant in which points are nearer to the origin then to the line x = 3.

6. Let $f(x) = \lim_{n\to\infty}\left(\frac{n!}{n^n}\right)^{\frac{1}{n}}$. Then the value of $[e^2 f(x)]$, where $[\bullet]$ denotes GIF.

Space for Rough work